

Chapter 15

Gaussian Processes in Machine Learning

15.1 Introduction

We have discussed in the previous chapters examples of supervised learning in the presence of a test set, such as the case of SVM, see chapter 14, and neural networks. Neural networks, for example, can be computationally heavy. Also, those ML models are not in general explicitly built within a Bayesian framework¹. Gaussian processes (GP) are appealing ML models, inherently Bayesian, and constitute another class of models in supervised learning. These models are based on the multi-normal distribution (briefly mentioned in chapter 1), which has useful properties. In standard parametric regression models, e.g. linear regression (chapter 5), the parameters (coefficients of the linear regression model) are estimated based on the training data. Within a Gaussian framework, as in the multiple linear regression, the coefficients are interpreted as random variables having mean values and variances. In this case we talk about probability distribution over variables, e.g. parameters, that is random variables. In GP, instead, we talk about probability distributions over functions. For example, a realization from a Gaussian RV is a number, but a realization of a GP is a function, i.e., a curve or path $f(x)$.

GP have some kind of overlap with SVM in the sense that the response variable is a combination of basis functions or kernels. Below we discuss the basis of GP and for more details the reader is referred to textbooks, e.g., Rasmussen and Williams (2006), Hsieh (2023), Bishop (2006), Lindholm et al. (2023), and Murphy (2012). We start first by outlining the properties of the multivariate Gaussian distribution, which constitutes the backbone of GP.

15.2 Properties of the multinormal Gaussian distribution

In GP we normally deal with multivariate normal probability distributions. We assume we have a d -dimensional normal variable X with mean μ and covariance matrix Σ . The

¹although BayesianNN are emerging to cover some of the limitations of conventional NN, see e.g., Arbel et al. (2023)

corresponding pdf $p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{|\boldsymbol{\Sigma}|}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right] \quad (15.1)$$

Note that sometimes we also use the notation $p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = N(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$. The Gaussian distributions are known to have very useful properties. For instance, the sum and also convolution of Gaussians is Gaussian, and the conditional distribution of Gaussian is also Gaussian. The something applies to the partition of Gaussian vectors. Precisely, if (X_1, X_2) is a partition of the Gaussian vector X and $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ are the respective means of X_1 and X_2 , and $\boldsymbol{\Sigma}_{11}$ and $\boldsymbol{\Sigma}_{22}$ the corresponding covariance matrices; so that $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are partitioned as follows:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{12}^T & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad (15.2)$$

then we know that X_1 and X_2 are also Gaussian. Equally important is the conditional distribution of $X_1|X_2$, which is also Gaussian with mean $\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$ and covariance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{12}^T$, that is:

$$p(X_1|X_2) \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{12}^T) \quad (15.3)$$

Remark

It is useful to remind here that given a sample of data $x_1, \dots, x_n$, and we look to fit a Gaussian distribution $p(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ to the sample. Using the likelihood function $L(\mu, \sigma^2) = \prod_{t=1}^n P(x_t|\mu, \sigma^2)$, then the sample mean $\bar{x}$ and the sample variance s^2 are respectively the maximum likelihood estimators of μ and σ^2 .

Next, assume we are given a marginal distribution $p(\mathbf{x}) \sim N(\mathbf{x}; \mathbf{m}, \mathbf{M})$ and a conditional distribution $p(\mathbf{y}|\mathbf{x}) \sim N(\mathbf{y}; \mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{N})$, and we want to find the joint distribution of $\mathbf{z} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$. The random vector $\mathbf{z}$ is multinormal, its pdf $p(\mathbf{z})$ can be obtained using Bayes rule, i.e. $\log p(\mathbf{z}) = \log p(\mathbf{x}) + \log p(\mathbf{y}|\mathbf{x})$. This last expression is quadratic meaning that $\mathbf{z}$ is indeed Gaussian. Its mean and covariance can be learned respectively from the linear and quadratic terms, and are given by:

$$\text{cov}(\mathbf{z}) = \begin{pmatrix} \mathbf{M} & \mathbf{M}\mathbf{A}^T \\ \mathbf{A}\mathbf{M} & \mathbf{N} + \mathbf{A}\mathbf{M}\mathbf{A}^T \end{pmatrix} \quad \text{and} \quad E(\mathbf{z}) = \begin{pmatrix} \mathbf{m} \\ \mathbf{A}\mathbf{m} + \mathbf{b} \end{pmatrix} \quad (15.4)$$

The marginal distribution $p(\mathbf{y})$ of $\mathbf{y}$ can be read directly from Eq 15.4. The conditional distribution $p(\mathbf{x}|\mathbf{y})$ can also be derived from Eq 15.4 (see also Eq 15.3), and denoting by $\mathbf{P} = (\mathbf{M}^{-1} + \mathbf{A}^T\mathbf{N}^{-1}\mathbf{A})^{-1}$, we get:

$$p(\mathbf{x}|\mathbf{y}) \sim N(\mathbf{x}; \mathbf{P}[\mathbf{A}^T\mathbf{N}^{-1}(\mathbf{y} - \mathbf{b}) + \mathbf{M}^{-1}\mathbf{m}], \mathbf{P}) \quad (15.5)$$

Exercise

Derive Eq 15.5 using Eq 15.3.

Hint. Let us look at the mean first. We use one form of Woodbury lemma $((\mathbf{A} + \mathbf{UCV}^T)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}^T\mathbf{A}^{-1})$. Also, we denote by $\mathbf{D} = \mathbf{M}^{-1} + \mathbf{A}'\mathbf{N}^{-1}\mathbf{A}$, where $\mathbf{A}' = \mathbf{A}^T$. The mean, following Eq. 15.3 is $\mathbf{m} - (\mathbf{MA}'\mathbf{N}^{-1}\mathbf{A} - \mathbf{MA}'\mathbf{N}^{-1}\mathbf{AD}^{-1}\mathbf{A}'\mathbf{N}^{-1}\mathbf{A})\mathbf{m} + (\mathbf{MA}'\mathbf{N}^{-1} - \mathbf{MA}'\mathbf{N}^{-1}\mathbf{AD}^{-1}\mathbf{A}'\mathbf{N}^{-1})(\mathbf{y} - \mathbf{b})$. In the coefficient of $\mathbf{m}$, insert $\mathbf{D}^{-1}\mathbf{D}$ after the first term $\mathbf{MA}'\mathbf{N}^{-1}\mathbf{A}$ and factorise. We end up with $\mathbf{m} - \mathbf{MA}'\mathbf{N}^{-1}\mathbf{AD}^{-1}\mathbf{M}^{-1}\mathbf{m}$. Then factorise out the term $(\mathbf{I} + \mathbf{MA}'\mathbf{N}^{-1}\mathbf{A})^{-1}$, to lead to $(\mathbf{I} + \mathbf{MA}'\mathbf{N}^{-1}\mathbf{A})^{-1}\mathbf{m}$. For the other term, just insert $\mathbf{DD}^{-1}$ between $\mathbf{M}$ and $\mathbf{A}'\mathbf{N}^{-1}$ then factorize by taking $\mathbf{D}^{-1}\mathbf{A}'\mathbf{N}^{-1}$ on the right hand side, to end up with $\mathbf{D}^{-1}\mathbf{A}'\mathbf{N}^{-1}(\mathbf{y} - \mathbf{b})$. Similar argument for the covariance, which is $\mathbf{M} - \mathbf{MA}'(\mathbf{N}^{-1} - \mathbf{N}^{-1}\mathbf{AD}^{-1}\mathbf{A}'\mathbf{N}^{-1})\mathbf{AM}$. Next insert $\mathbf{DD}^{-1}$ between $\mathbf{M}$ and $\mathbf{A}'\mathbf{N}^{-1}$ on the left hand side, then factorise out the term $\mathbf{D}^{-1}\mathbf{A}'\mathbf{N}^{-1}$ on the right hand side. This yields $\mathbf{M} - \mathbf{D}^{-1}\mathbf{A}'\mathbf{N}^{-1}\mathbf{AM}$. Finally insert again $\mathbf{D}^{-1}\mathbf{D}$ on the left hand side of the first $\mathbf{M}$. This finally yields $\mathbf{D}^{-1}$, and hence Eqs. 15.5 and 15.3 are equivalent.

15.3 Gaussian processes

15.3.1 Random function

As it is mentioned in the introduction GP involves distributions over functions rather than distributions over variables. Since we talk about distribution over functions, and before delving into GP, let us talk about the concept of random functions. Consider the following "function" defined by:

$$F(x, \omega) = e^{-\omega x^2} \cos(\pi \omega x + 2\omega) \quad (15.6)$$

where ω is a random variable, e.g., normal $N(0, 1)$, and x is any real number within $[-1, 1]$.

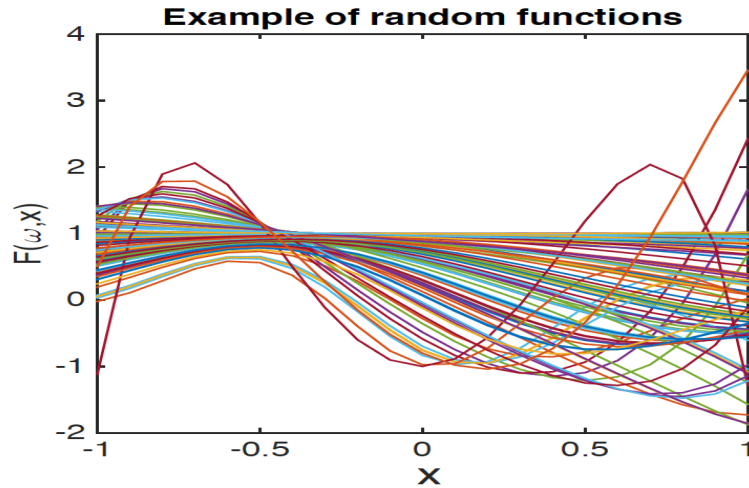


Figure 15.1: Example of realizations of a random function given by Eq. 15.6

For a given x_0 in $[-1, 1]$, $F(x_0, \omega)$ is a random variable. But for any realization ω_0 from $N(0, 1)$, $F(x, \omega_0)$ becomes a real function defined over the interval $[-1, 1]$. Figure 15.1 shows an example of realizations of a random function defined by Eq. 15.6 with $\omega = \frac{1}{2} + \frac{3}{5}N(0, 1)$.

The above random function is a simple example of a stochastic process. In the theory of probability the concept of stochastic (or random) process refers to a family of random

variables, in a given probability space, indexed by a set that can be discrete (like integers) or continuous (like real numbers), and can be infinite. The Markov process or AR(1) model, discussed in chapter 2, is an example.

15.3.2 Gaussian processes

In the same vein a Gaussian process is a set or collection of random variables for which any finite subset constitutes a vector of multivariate Gaussian distribution.

Remark

In general if the index is integer or real we talk about process, and if it is higher dimension, like a subset of $\mathbb{R}^2$ we talk about fields.

A GP, noted by $f \sim GP(m, K)$ is normally defined by a mean function $m(x)$ and covariance function $K(x, x')$. The latter is a symmetric positive semi-definite function. This means that for any numbers $x_1, \dots, x_n$ and $a_1, \dots, a_n$ we have:

$$\sum_{i,j=1}^n a_i K(x_i, x_j) a_j \geq 0 \quad (15.7)$$

Eq. 15.7 means that the symmetric matrix $\mathbf{K} = (K_{ij})$, with $K_{ij} = K(x_i, x_j)$ is a covariance matrix. Given any training set $x_1, \dots, x_n$, the functions $f(x_1), \dots, f(x_n)$, of the Gaussian process $f \sim GP(m, K)$, are random variables. The covariance between $f(x_i)$ and $f(x_j)$ is precisely $K(x_i, x_j)$:

$$K(x_i, x_j) = \text{cov}(f(x_i), f(x_j)) = E[(f(x_i) - m(x_i))(f(x_j) - m(x_j))] \quad (15.8)$$

Example 1: AR(1)

The Markov process $x_{t+1} = \alpha x_t + \varepsilon_{t+1}$, where ε_t are iid $N(0, \sigma^2)$, is a simple example of a (discrete) Gaussian process. The mean function is the zero function and $K(x, x')$ is provided by its (stationary) autocovariance function (see chapter 2).

Example 2

Let $m(x) = \frac{1}{10}x^3$ and $K(x, x') = \frac{1}{\sqrt{4\pi}}e^{-(x-x')^2/4}$. Then $f \sim GP(m, K)$ is a Gaussian process. For any integer n and numbers $x_1, \dots, x_n$, we obtain a multivariate Gaussian distribution $N(\mathbf{m}, \mathbf{K})$ with mean $\mathbf{m} = (m(x_1), \dots, m(x_n))^T$ and covariance matrix $\mathbf{K} = (K_{ij})$, with $K_{ij} = K(x_i, x_j)$. Figure 15.2 displays an example showing 30 curves (or paths) generated from $GP(m, K)$ defined in the interval $[-2, 2]$. These paths are constructed as follows. First, choose a regular grid of $[-2, 2]$ using for example a step of 0.1 and obtain an array of length $n = 41$, $x_i = -2 + (i-1)*0.1$, for $i = 1, \dots, 41$, then calculate $\boldsymbol{\mu}$ and $\mathbf{K}$. Finally 30 realizations are generated from $N(\boldsymbol{\mu}, \mathbf{K})$; each realization is a vector $\mathbf{y}$ and constitutes one path.

Note that in the literature of GP, the above is referred to as prior GP since there is no data involved. Once observed data become available one can delve into regression (or classification) using posterior GP as discussed next.

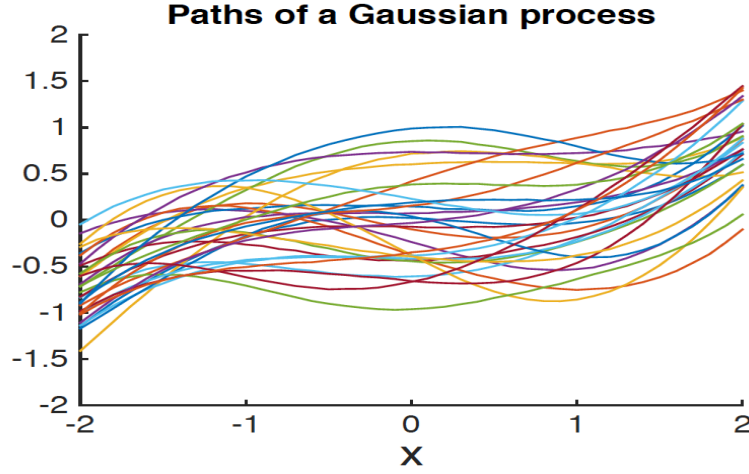


Figure 15.2: Examples of 30 paths generated with a $GP(m, K)$ described in Example 2 above.

Remark

A GP can also be defined using random function:

$$f(\omega, x) = \sum w_i \phi_i(x) + m(x) \quad (15.9)$$

with $w_i \sim N(0, \lambda_i)$ (iid) and $m(x)$ is a mean function. Note that the sum in Eq 15.9 can be finite or infinite. An interesting case emerges when the functions $\phi_i()$ derive from a kernel $K()$ (Boser et al. 1992, Schölkopf et al. 1998). Using the kernel trick the covariance function $cov(f(\omega, x), f(\omega, y)) = \sum_{i,j} E(w_i, w_j) \phi_i(x) \phi_j(y)$, becomes $\sum_i \lambda_i \phi_i(x) \phi_i(y) = K(x, y)$. Here λ_i and $\phi_i()$ are, respectively, the i 'th eigenvalue and eigenvector of the kernel operator $K()$.

15.4 GP regression

As it is mentioned in the introduction the GP regression is embedded naturally within a Bayesian framework, and we briefly discuss first the Bayesian linear regression to set up the scene for GP regression.

15.4.1 Introduction to Bayesian linear regression

Given observations $(\mathbf{x}_i, y_i)$, composed of input $\mathbf{x}_i$ and response y_i , $i = 1, \dots, n$, the standard multiple linear regression is discussed in chapter 16, and reads:

$$y = \mathbf{a}^T \mathbf{x} + \varepsilon \quad (15.10)$$

where $\varepsilon \sim N(0, \sigma^2)$. The probability distribution of y , given parameter $\mathbf{a}$ is normal with mean $\mathbf{a}^T \mathbf{x}$ and variance σ^2 , i.e. $p(y|\mathbf{a}) = N(y; \mathbf{a}^T \mathbf{x}, \sigma^2)$. Designating by $\mathbf{X}$ the design matrix, and letting $\mathbf{y} = (y_1, \dots, y_n)^T$, the above conditional distribution (based on independence) becomes:

$$p(\mathbf{y}|\mathbf{a}) = N(\mathbf{y}; \mathbf{X}\mathbf{a}, \sigma^2 \mathbf{I}) \quad (15.11)$$

which equals to the product of $p(y_i|\mathbf{a})$, $i = 1, \dots, n$.

In a Bayesian setting the joint distribution $p(\mathbf{y}, \mathbf{a})$ is factorised using the Bayesian rule (see chapter 1) as $p(\mathbf{y}, \mathbf{a}) = p(\mathbf{y}|\mathbf{a})p(\mathbf{a}) = p(\mathbf{a}|\mathbf{y})p(\mathbf{y})$, from which one gets the posterior distribution expressing our updated knowledge of the parameters upon obtaining new data:

$$p(\mathbf{a}|\mathbf{y}) = \frac{1}{p(\mathbf{y})}p(\mathbf{y}|\mathbf{a})p(\mathbf{a}) \quad (15.12)$$

In Eq 15.12 the denominator is a normalization factor and therefore we can re-write it as $p(\mathbf{a}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{a})p(\mathbf{a})$. The term $p(\mathbf{y}|\mathbf{a})$ is the likelihood, and $p(\mathbf{a})$ is the prior distribution of the regression parameter $\mathbf{a}$. This prior is the belief we hold about parameter $\mathbf{a}$ before any observational data is obtained. Once we get new observations our belief will be updated based on Eq 15.12. For a Gaussian prior $p(\mathbf{a}) = N(\mathbf{a}; \mathbf{a}_0, \sigma_0^2 \mathbf{I})$, along with Eq 15.11, the posterior is also Gaussian, i.e.,

$$p(\mathbf{a}|\mathbf{y}) = N(\mathbf{a}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (15.13)$$

with $\boldsymbol{\Sigma} = (\sigma_0^{-2} \mathbf{I} + \sigma^{-2} \mathbf{X}^T \mathbf{X})^{-1}$ and $\boldsymbol{\mu} = \boldsymbol{\Sigma}(\sigma_0^{-2} \mathbf{a}_0 + \sigma^{-2} \mathbf{X}^T \mathbf{y})$. Note that this posterior becomes non-Gaussian if the prior is not Gaussian, e.g. uniform.

Exercise

Derive Eq 15.13 (use results from section 2).

Once the posterior distribution is learned using the training data $(\mathbf{X}, \mathbf{y})$, the next step is to use the model for prediction. Give a test point $\mathbf{x}_*$, the objective (in the Bayesian context) is to obtain $p(y_*|\mathbf{y})$. This is readily obtained using the marginal distribution of $p(y_*, \mathbf{a}|\mathbf{y})$, i.e.,

$$p(y_*|\mathbf{y}) = \int p(y_*|\mathbf{a})p(\mathbf{a}|\mathbf{y})d\mathbf{a} \quad (15.14)$$

which is referred to as the predictive posterior distribution. In this equation, $p(\mathbf{a}|\mathbf{y})$ is given by Eq 15.13, i.e. $N(\mathbf{a}^T \mathbf{x}_*, \boldsymbol{\Sigma})$. The predictive posterior of $\mathbf{a}^T \mathbf{x}_*$ is obtained directly from Eq 15.13:

$$p(\mathbf{a}^T \mathbf{x}_*|\mathbf{y}) = N(\mathbf{a}^T \mathbf{x}_*; \mathbf{x}_*^T \boldsymbol{\mu}, \mathbf{x}_*^T \boldsymbol{\Sigma} \mathbf{x}_*). \quad (15.15)$$

Eq 15.10 also yields the posterior distribution of $\mathbf{y}_*$, i.e., $p(\mathbf{y}_*|\mathbf{y}) = N(\mathbf{y}_*; \mathbf{x}_*^T \boldsymbol{\mu}, \mathbf{x}_*^T \boldsymbol{\Sigma} \mathbf{x}_* + \sigma^2)$. Note that the term $p(\mathbf{y}|\mathbf{a})$ is the likelihood and $p(\mathbf{y})$ is the marginal likelihood or evidence.

Remarks

- The parameter σ^2 in Eq 15.10 and the parameters of the posterior distribution can be estimated by maximizing the marginal log-likelihood $L = \log p(\mathbf{y}) = -\frac{1}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$
- The above model, Eq 15.10, can be extended using a set of basis functions $y = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$, where $\mathbf{w}$ is the weight vector, Gaussian distributed with prior $N(\mathbf{0}, \sigma^2 \mathbf{I})$, and $\boldsymbol{\phi} = (\phi_1, \dots, \phi_M)^T$. This leads to distribution over functions $y(\mathbf{x})$ (random function) with mean zero and covariance function $\sigma^2 \boldsymbol{\phi}^T(\mathbf{x}_i) \boldsymbol{\phi}(\mathbf{x}_j) = K(\mathbf{x}_i, \mathbf{x}_j)$, with $K()$ being the kernel function.

15.4.2 Gaussian process regression

Now we assume we have a GP defined by its mean function $m(\mathbf{x})$ and covariance function $K(\mathbf{x}_i, \mathbf{x}_j)$. This is the prior distribution over function, which is used for Bayesian inference. Next, we will use this framework to obtain the posterior distribution when we are given a set of function values $\mathbf{f} = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T$, assumed noise-free, observed at a given training data $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$, i.e., observations $y_i = f(\mathbf{x}_i)$, $i = 1, \dots, n$. The regression function is:

$$p(\mathbf{y}|\mathbf{X}) = p(\mathbf{f}|\mathbf{X}) \sim N(\mathbf{f}; \mathbf{m}, \mathbf{K}) \quad (15.16)$$

where $\mathbf{m} = (m(\mathbf{x}_1), \dots, m(\mathbf{x}_n))^T$, and $\mathbf{K} = (K_{ij})$ where $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j)$. The matrix $\mathbf{K}$ is also denoted by $\mathbf{K} = K(\mathbf{X}, \mathbf{X})$. In general it is standard to take a zero-mean function, $\boldsymbol{\mu} = \mathbf{0}$ for the prior.

The posterior distribution can also be used for prediction. Given a new test input $\mathbf{x}_*$, the objective is to estimate the predictive posterior of $\mathbf{y}_* = f(\mathbf{x}_*) = f_*$, i.e., $p(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*)$. This can be obtained from the (Gaussian) joint distribution $p(\mathbf{f}, \mathbf{f}_*|\mathbf{X}, \mathbf{x}_*)$:

$$\begin{pmatrix} \mathbf{f} \\ f_* \end{pmatrix} \sim N\left(\begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{K} & \mathbf{k}_* \\ \mathbf{k}_*^T & k_{**} \end{bmatrix}\right) \quad (15.17)$$

where $\mathbf{k}_* = K(\mathbf{X}, \mathbf{x}_*)$, i.e. $\mathbf{k}_* = (K(\mathbf{x}_1, \mathbf{x}_*), \dots, K(\mathbf{x}_n, \mathbf{x}_*))^T$, and $k_{**} = K(\mathbf{x}_*, \mathbf{x}_*)$. The predictive posterior is obtained using results from section 2:

$$p(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) \sim N(\mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f}, k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*) \quad (15.18)$$

In case of noisy observations, $y_i = f(\mathbf{x}_i) + \varepsilon_i$, where ε_i are iid $N(0, \sigma^2)$ and $f \sim GP(0, K)$. In this case the covariance of $\mathbf{y} = (y_1, \dots, y_n)^T$ is $cov(\mathbf{y}) = \mathbf{K} + \sigma^2 \mathbf{I}$, leading to the posterior:

$$p(\mathbf{y}|\mathbf{X}) \sim N(\mathbf{y}; \mathbf{0}, \mathbf{K} + \sigma^2 \mathbf{I}) \quad (15.19)$$

and the predictive posterior, given input test $\mathbf{x}_*$, becomes:

$$p(f_*|\mathbf{y}, \mathbf{X}, \mathbf{x}_*) \sim N(\mathbf{k}_*^T (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}, k_{**} - \mathbf{k}_*^T (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_*) \quad (15.20)$$

Remark

Eq 15.20 shows that the GP prediction uncertainty $k_{**} - \mathbf{k}_*^T (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_*$ depends on the inputs $\mathbf{X}$ and $\mathbf{x}_*$ and not on the observations $\mathbf{y}$.

The above formulation can be summarized in a few lines algorithm. We start first by applying Cholesky decomposition $\mathbf{K} + \sigma^2 \mathbf{I} = \mathbf{C}\mathbf{C}^T$, then get the mean and variance:

$$\begin{aligned} \mathbf{u} &= \mathbf{C}^{-T} \mathbf{C}^{-1} \mathbf{y} \\ y_* &= \mathbf{k}_*^T \mathbf{u} \text{ \& } \mathbf{v} = \mathbf{C}^{-1} \mathbf{k}_* \\ cov(y_*) &= k_{**} - \mathbf{v}^T \mathbf{v} \end{aligned}$$

The marginal log-likelihood is derived from Eq 15.19 and reads (up to a constant):

$$L = -\frac{1}{2} \mathbf{y}^T (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K} + \sigma^2 \mathbf{I}| \quad (15.21)$$

15.5 Gaussian process classification

Before discussing the GP classification, let us go back to the concept of logistic regression, briefly discussed in the previous chapters, but with more link to the classification problem in the context of Gaussian processes.

15.5.1 Logistic regression and classification

Logistic regression

Logistic regression corresponds to the case where the outputs are categorical, with continuous inputs. It is a classification problem and constitutes a particular case of GLM (see chapter 11). Consider the binary case first, with output $y \in \{0, 1\}$. The data set comprises $\{\phi(\mathbf{x}_t), y_t\}$, $t = 1, \dots, n$, and $\phi()$ is a transformation from the input space into the feature space. Note that the standard logistic regression corresponds to a linear function $\phi()$, but for the sake of generality we use the notation $\phi()$, which can be linear or nonlinear. The posterior probability of, say, class $\mathcal{C}_1$, for a given input $\phi(\mathbf{x})$, can be expressed via the logistic function:

$$p(\mathcal{C}_1|\phi(\mathbf{x})) = \sigma(\mathbf{w}^T \phi(\mathbf{x})) \quad (15.22)$$

where $\sigma(x) = (1 + \exp(-x))^{-1}$ is the logistic sigmoid function. Denoting by $z = \mathbf{w}^T \phi(\mathbf{x})$, the binary case corresponds to a Bernoulli distribution, i.e.,

$$p(y|z) = (\sigma(z))^y (1 - \sigma(z))^{1-y} \quad (15.23)$$

To estimate the weight $\mathbf{w}$, we maximize the likelihood² $L(\mathbf{w})$:

$$L(\mathbf{w}) = p(\mathbf{y}|\mathbf{w}) = \prod_{i=1}^n [\sigma(z_i)]^{y_i} [1 - \sigma(z_i)]^{1-y_i} \quad (15.24)$$

where $z_i = \mathbf{w}^T \phi(\mathbf{x}_i)$, $i = 1, \dots, n$. It is easier to maximize the log-likelihood, whose gradient is:

$$\nabla \log(L(\mathbf{w})) = \sum_{i=1}^n (\sigma(z_i) - y_i) \phi(\mathbf{x}_i) \quad (15.25)$$

The log-likelihood, being non-quadratic, is optimized using an iterative Newton scheme making use of the Hessian matrix, and yields point estimates of the weight $\mathbf{w}$. The probability of the class of a new input data is then obtained using Eq 15.22.

Multinomial regression

the standard multinomial logistic, or multinomial logit (Mlogit) regression model is an extension of the standard logistic (binomial) regression, and we discuss it briefly here, which will be used later in GP classification. We assume we have C categories ($C > 2$) in which a membership probability is compared to a reference category (taken here to be the first category). The Mlogit regression model predicts the probability of different possible outcomes

²or minimizing its inverse.

of categorical dependent random variables given independent variables. The log-odd ratio³ for $c = 2, \dots, C$ is defined (see also chapter 12, section 3) as:

$$\log \frac{Pr(Y = c)}{Pr(Y = 1)} = b_{0c} + b_{1c}x_1 + \dots + b_{Mc}x_M = \mathbf{b}_c^T \mathbf{x} \quad (15.26)$$

Equation 15.26 leads to:

$$Pr(Y = c) = Pr(Y = 1)e^{\mathbf{b}_c^T \mathbf{x}}, \quad c = 2, \dots, C \quad (15.27)$$

Summing Eq 15.27 for $c = 2, \dots, C$ yields $Pr(Y = 1) = (1 + \sum_{l=2}^C e^{\mathbf{b}_l^T \mathbf{x}})^{-1}$, the model then becomes:

$$Pr(Y = c|\mathbf{x}) = \frac{e^{\mathbf{b}_c^T \mathbf{x}}}{1 + \sum_{l=2}^C e^{\mathbf{b}_l^T \mathbf{x}}}, \quad c = 2, \dots, C \quad (15.28)$$

Remark

- Note that by applying Eq 15.27 to $c = 1$, one gets $e^{\mathbf{b}_1^T \mathbf{x}} = 1$, Eq 15.28 can then be written as:

$$Pr(Y = c|\mathbf{x}) = \frac{e^{\mathbf{b}_c^T \mathbf{x}}}{\sum_{l=1}^C e^{\mathbf{b}_l^T \mathbf{x}}}, \quad c = 1, \dots, C \quad (15.29)$$

and highlights the importance of the softmax function, which links membership probabilities to linear predictors.

- The application of Eq 15.28 with $C = 2$, we get $P(Y = 2|\mathbf{x}) = \sigma(\mathbf{b}_2^T \mathbf{x})$, with $\sigma()$ being the logistic function, and then $P(Y = 2|\mathbf{x}) = 1 - P(Y = 1|\mathbf{x}) = \sigma(-\mathbf{b}_2^T \mathbf{x})$, which is the precisley the (standard) logistic regression model.

The standard Mlogit can be generalized by using a nonlinear transformation $\phi()$. In this case the posterior probability of the c 'th class $\mathcal{C}_c$ is similar to Eq 15.29, and is given by the softmax function:

$$p(\mathcal{C}_c|\phi(\mathbf{x})) = \frac{e^{\mathbf{w}_c^T \phi(\mathbf{x})}}{\sum_l e^{\mathbf{w}_l^T \phi(\mathbf{x})}} \quad (15.30)$$

The remaining steps are similar to the binary case, and it is expanded later in the section on multi-class case.

Bayesian logistic regression

In the context of GP classification, the Bayesian approach is the focal point. In Bayesian logistic regression the posterior predictive distribution cannot be obtained analytically in general as it involves a convolution of non-Gaussian distributions.. One way is to use Monte-Carlo calculation. The standard and more common method is to use the Laplace approximation, which is also discussed later in section 5, and we discuss here the binary case, see next section for the multi-class case. Following this approximation one seeks a Gaussian approximation of a non-Gaussian distribution. We assume that the weight $\mathbf{w}$ prior is normal, i.e., $p(\mathbf{w}) \sim N(\mathbf{a}_0, \mathbf{S}_0)$, and the posterior is given by Bayes law:

$$p(\mathbf{w}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{w})p(\mathbf{w}) \quad (15.31)$$

³Generally defined as the ratio of odds of success to the odds of failure

where $p(\mathbf{y}|\mathbf{w})$ is the likelihood, Eq 15.24, and $\mathbf{y} = (y_1, \dots, y_n)$. As with the Bayesian linear regression, see Eq 15.15, the objective is to obtain the predictive distribution of class $\mathcal{C}_1$, given a new input $\phi(\mathbf{x})$, which can be obtained by marginalizing over $\mathbf{w}$:

$$p(\mathcal{C}_1|\mathbf{y}) = \int p(\mathcal{C}_1|\mathbf{w})p(\mathbf{w}|\mathbf{y})d\mathbf{w} \quad (15.32)$$

With Laplace approximation we seek to approximate the posterior $p(\mathbf{w}|\mathbf{y})$ in Eq 15.31 by a Gaussian. This can be achieved by approximating $\log p(\mathbf{w}|\mathbf{y})$ (up to a constant) by a quadratic expression:

$$\begin{aligned} \log(p(\mathbf{w}|\mathbf{y})) &= -\frac{1}{2}(\mathbf{w} - \mathbf{a}_0)^T \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{a}_0) + \sum_{i=1}^n y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \\ &\approx -\frac{1}{2}(\mathbf{w} - \mathbf{a}_1)^T \mathbf{S}_1^{-1}(\mathbf{w} - \mathbf{a}_1) \end{aligned} \quad (15.33)$$

where $\mathbf{a}_1$ and $\mathbf{S}_1^{-1}$ are given, respectively, by the maximum and the opposite Hessian of $\log p(\mathbf{w}|\mathbf{y})$ resulting from the optimization. We will also apply Laplace approximation below in connection with GP classification.

Once the posterior is approximated by a Gaussian following Eq 15.33, the predictive distribution is approximated by:

$$p(\mathcal{C}_1|\mathbf{y}) \approx \int \sigma(\mathbf{w}^T \phi(\mathbf{x})) N(\mathbf{w}; \mathbf{a}_1, \mathbf{S}_1) d\mathbf{w} = \int \sigma(z) N(z; \mathbf{a}_1^T \phi(\mathbf{x}), \phi(\mathbf{x})^T \mathbf{S}_1 \phi(\mathbf{x})) dz \quad (15.34)$$

The last integral cannot be computed analytically. However, by approximating the sigmoid $\sigma()$ by the inverse probit, i.e. the normal cdf function $\Phi()$, the integral can be computed analytically (see, e.g., Spiegelhalter and Lauritzen (1990), or Bishop (2006) and references therein), and results in the following identity:

$$\int \Phi(z\sqrt{\pi/8}) N(z; \mu, \sigma^2) dz = \Phi\left(\frac{\mu}{\sqrt{\sigma^2 + 8/\pi}}\right) \quad (15.35)$$

which can be used to compute the convolution of a normal with a sigmoid, i.e.,

$$\int \sigma(z) N(z; \mu, \sigma^2) dz \approx \sigma\left(\frac{\mu}{\sqrt{1 + \pi\sigma^2/8}}\right) \quad (15.36)$$

Note that in the above expression, $\sigma(z)$ is approximated by $\Phi(z\sqrt{\pi/8})$. Finally, the predictive posterior boils down to:

$$p(\mathcal{C}_1|\phi(\mathbf{x}), \mathbf{y}) \approx \sigma\left(\frac{\mathbf{m}_1^T \phi(\mathbf{x})}{\sqrt{1 + \pi\phi(\mathbf{x})^T \mathbf{S}_1 \phi(\mathbf{x})/8}}\right) \quad (15.37)$$

15.5.2 GP classification: the binary case

Gaussian processes can also be used in classification. However, compared to regression the posterior distribution for classification is not Gaussian because the response is categorical as in the case of logistic regression. In this case few approximations can be used to obtain Gaussian predictive distribution, and the most common is Laplace approximation, which aims in essence at approximating a given distribution by a Gaussian as discussed in the previous subsection, and we basically follow the same steps. We consider the binary case in which the response of the training sample y is in $\{0, 1\}$. The input training sample is $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$

with corresponding output $\mathbf{y} = (y_1, \dots, y_n)^T$ in $\{0, 1\}$. We also assume we have a test point $\mathbf{x}_*$ for which we want to predict its class y_* . Since the GP spans the set of real numbers, a logistic mapping is applied to the GP $f(\mathbf{x})$, i.e., $\sigma(f(\mathbf{x})) = (1 + \exp(-f(\mathbf{x})))^{-1} \in [0, 1]$, which can be used to decide on the probability of belonging to either class $\mathcal{C}_0$ or $\mathcal{C}_1$. In this binary case we are dealing with Bernoulli distribution (see also Eq 15.23):

$$p(y|\sigma(f(\mathbf{x}))) = [\sigma(f(\mathbf{x}))]^y [1 - \sigma(f(\mathbf{x}))]^{1-y} \quad (15.38)$$

The objective is to estimate the predictive distribution $p(y_*|\mathbf{y}, \mathbf{X}, \mathbf{x}_*)$. We consider the joint vector $(\mathbf{y}^T, y_*)$ and a Gaussian process prior $f \sim GP(0, K)$ defined by a covariance function $K()$. Letting $\mathbf{f} = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T$ and $f_* = f(\mathbf{x}_*)$, the joint distribution of $(\mathbf{f}^T, f_*)$ is given by Eq 15.17. Using Eq 15.38, the predictive probability $p(y_* = 1|\mathbf{y})$ can be obtained using marginalisation as:

$$\int p(y_*, f_*|\mathbf{y})df_* = \int p(y_*|f_*)p(f_*|\mathbf{y})df_* = \int \sigma(f(\mathbf{x}_*))p(f_*|\mathbf{y})df_* \quad (15.39)$$

Note that in the above equation we dropped $\mathbf{X}$ and $\mathbf{x}_*$ from $p(y_*|\mathbf{y}, \mathbf{X}, \mathbf{x}_*)$ just for notational purpose. The next step is to use the Laplace approximation applied to $p(f_*|\mathbf{y})$. The starting point is to marginalise it with respect to $\mathbf{f} = (f_1, \dots, f_n)^T$, i.e. $\int p(\mathbf{f}, f_*|\mathbf{y})d\mathbf{f}$, and then use Bayes' rule applied to $p(\mathbf{f}, f_*|\mathbf{y})$, that is $p(\mathbf{f}, f_*|\mathbf{y}) \propto p(\mathbf{f}, f_*)p(\mathbf{y}|\mathbf{f}, f_*) = p(\mathbf{f}, f_*)p(\mathbf{y}|\mathbf{f})$. This yields:

$$p(f_*|\mathbf{y}) = \int p(f_*|\mathbf{f})p(\mathbf{f}|\mathbf{y})d\mathbf{f} \quad (15.40)$$

In Eq 15.40, $p(f_*|\mathbf{f})$ is Gaussian $N(f_*; \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f}, k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*)$, following Eq 15.17. Lastly we need to gaussianise the likelihood $p(\mathbf{f}|\mathbf{y})$. Recall now that

$$p(\mathbf{y}|\mathbf{f}) = \prod_{t=1}^n [\sigma(f(\mathbf{x}_t))]^{y_t} [1 - \sigma(f(\mathbf{x}_t))]^{1-y_t} = \prod_{t=1}^n \left(e^{f_t y_t} \sigma(-f_t) \right) \quad (15.41)$$

Now keeping in mind that $p(\mathbf{f}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{f})p(\mathbf{f})$, one gets (up to a constant):

$$\log p(\mathbf{f}|\mathbf{y}) = L(\mathbf{f}) = -\frac{1}{2} \mathbf{f}^T \mathbf{K}^{-1} \mathbf{f} - \frac{1}{2} \log |\mathbf{K}| + \mathbf{y}^T \mathbf{f} - \log \prod_{m=1}^n (1 + e^{f_m}) \quad (15.42)$$

As with Bayesian logistic regression, in order for $p(\mathbf{f}|\mathbf{y})$ to be approximated by a Gaussian Eq 15.42 must "look like" a quadratic. Its gradient and Hessian are respectively:

$$\begin{aligned} \mathbf{g} &= \nabla L(\mathbf{f}) = \mathbf{y} - \mathbf{K}^{-1} \mathbf{f} - \mathbf{w} \\ \mathbf{H} &= -\mathbf{K}^{-1} - \mathbf{W} = -(\mathbf{W}\mathbf{K} + \mathbf{I})\mathbf{K}^{-1} \end{aligned} \quad (15.43)$$

where $\mathbf{w} = (\sigma(f_1), \dots, \sigma(f_n))^T$, and $\mathbf{W} = \text{diag}(\sigma(f_1)\sigma(-f_1), \dots, \sigma(f_n)\sigma(-f_n))$, and $\text{diag}()$ is the diagonal operator. Since these are nonlinear in $\mathbf{f}$ the mode $\nabla L(\mathbf{f}) = \mathbf{0}$ is sought by using an iterative scheme such as Newton-Raphson (e.g., Bishop 2006, Wang 2026):

$$\mathbf{f}^{(k+1)} = \mathbf{f}^{(k)} - \mathbf{H}^{-1} \mathbf{g}(\mathbf{f}^{(k)}) \quad (15.44)$$

Note that the Hessian is negative definite and $L(\mathbf{f})$, being concave, has a unique maximum, and Eq 15.44 converges to a unique mode $\hat{\mathbf{f}}$ satisfying $\hat{\mathbf{f}} = \mathbf{K}(\mathbf{y} - \mathbf{w})$, and we get:

$$p(\mathbf{f}|\mathbf{y}) \sim N(\hat{\mathbf{f}}, -\mathbf{H}^{-1}) \quad (15.45)$$

in which $\mathbf{H}$, see Eq 15.43, and $\mathbf{w}$ are evaluated at $\hat{\mathbf{f}}$. Now that $p(\mathbf{f}|\mathbf{y})$ is approximated by a Gaussian as shown in Eq 15.45, the integral in Eq 15.40, which looks like a convolution of Gaussians, is also Gaussian. It can be computed, and after some algebra one gets:

$$\begin{aligned} p(f_*|\mathbf{y}) &\approx \int N(f_*; \mathbf{k}_* \mathbf{K}^{-1} \mathbf{f}, k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f}) N(\mathbf{f}; \hat{\mathbf{f}}, -\mathbf{H}^{-1}) d\mathbf{f} \\ &= N(f_*; \mathbf{k}_*^T (\mathbf{y} - \mathbf{w}), k_{**} - \mathbf{k}_*^T (\mathbf{W}^{-1} + \mathbf{K})^{-1} \mathbf{k}_*) \end{aligned} \quad (15.46)$$

Remark

The evaluation of Eq 15.46 is lengthy, but since it is Gaussian we can simply evaluate its mean and variance using marginalisation of the expectation and the iterated variance, see e.g., Murphy (2012), Rasmussen and Williams (2006). Recalling that $E(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) = \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f}$ (see section 4), then $E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y})$ is equal to $\int E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{f}) p(\mathbf{f}|\mathbf{X}, \mathbf{y}) d\mathbf{f}$, that is $\mathbf{k}_*^T \mathbf{K}^{-1} E(\mathbf{f}|\mathbf{X}, \mathbf{y}) \approx \mathbf{k}_*^T \mathbf{K}^{-1} \hat{\mathbf{f}}$ (see Eq 15.45).

The iterated variance⁴, using again results from GP regression (section 4), i.e. $var(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*, \mathbf{y}) = var(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) = k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*$, also, $E(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*, \mathbf{y}) = E(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) = \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f}$. These two identities yield $var(f_*) = E(k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*) + var(\mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{f})$. Taking $cov(\mathbf{f}|\mathbf{X}, \mathbf{x}_*, \mathbf{y})$ from Eq 15.45, the iterated variance, $var(f_*) = k_{**} - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_* + \mathbf{k}_*^T \mathbf{K}^{-1} cov(\mathbf{f}|\mathbf{y}) \mathbf{K}^{-1} \mathbf{k}_*$ finally leads to:

$$\begin{aligned} var(f_*) &= k_{**} - \mathbf{k}_*^T [\mathbf{K}^{-1} - \mathbf{K}^{-1} (\mathbf{K}^{-1} + \mathbf{W})^{-1} \mathbf{K}^{-1}] \mathbf{k}_* \\ &= k_{**} - \mathbf{k}_*^T (\mathbf{W}^{-1} + \mathbf{K})^{-1} \mathbf{k}_* \end{aligned} \quad (15.47)$$

where the Woodbury matrix identity⁵ was used. This finally yields

$$p(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) \sim N(f_*; \mathbf{k}_*^T \mathbf{K}^{-1} \hat{\mathbf{f}}, k_{**} - \mathbf{k}_*^T (\mathbf{W}^{-1} + \mathbf{K})^{-1} \mathbf{k}_*) \quad (15.48)$$

Exercise

In the above remark, the following identity was used; $E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) = \int E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{f}) p(\mathbf{f}|\mathbf{X}, \mathbf{y}) d\mathbf{f}$. Derive this identity.

Hint. By definition, $E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) = \int f_* p(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) df_*$. Then write $p(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y})$ as a marginalization over $\mathbf{f}$, that is, $\int p(f_*, \mathbf{f}|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) d\mathbf{f} = \int p(\mathbf{f}|\mathbf{X}, \mathbf{y}) p(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) d\mathbf{f}$. Using this in the definition of $E(f_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y})$, and after arrangement of the terms in the integral we get $\int p(\mathbf{f}|\mathbf{X}, \mathbf{y}) d\mathbf{f} [\int f_* p(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) df_*]$. This is precisely $\int E(f_*|\mathbf{f}, \mathbf{X}, \mathbf{x}_*) p(\mathbf{f}|\mathbf{X}, \mathbf{y}) d\mathbf{f}$.

Finally, one can now get back to Eq 15.39 to evaluate the binary prediction $\pi_* = P(y_*|\mathbf{X}, \mathbf{x}_*, \mathbf{y})$. As it has been pointed out in the previous sub-section, this integral, being a convolution of a Gaussian and a sigmoid, is theoretically intractable, and to compute it one can use the "probit" approximation in which the logistic sigmoid $\sigma()$ is approximated by the inverse probit function $\Phi()$, see Eq 15.35. Following this approximation,

⁴ $var(X) = E(var(X|Y)) + var(E(X|Y))$
⁵ $(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}(\mathbf{A}\mathbf{B}^{-1} + \mathbf{I})^{-1}$

$\int \sigma(x)N(x; \mu, \sigma^2)dx \approx \sigma(\frac{\mu}{\sqrt{1+\pi\sigma^2/8}})$. Finally one gets:

$$\pi_* = p(y_* = 1|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) \approx \sigma\left(\frac{E(f_*)}{\sqrt{1+\pi\text{var}(f_*)/8}}\right) \quad (15.49)$$

where $E(f_*)$ and $\text{var}(f_*)$ are obtained from Eq 15.46. Note that the decision boundary between the two classes is obtained by identifying the curve $p(y_* = 1|\mathbf{X}, \mathbf{x}_*, \mathbf{y}) = 0.5$.

Remark

If the class labels are $\{1, -1\}$ instead of $\{0, 1\}$ then the probability of belonging to one of the two classes can be written as⁶ $p(y|f) = \sigma(yf)$, and for an iid sample $p(\mathbf{y}|\mathbf{f}) = \prod_{i=1}^n \sigma(y_i f_i)$. The posterior distribution, Eq 15.45, becomes $p(\mathbf{f}|\mathbf{X}, \mathbf{y}) \approx N(\mathbf{K}\mathbf{w}, (\mathbf{K}^{-1} + \mathbf{W})^{-1})$, where $\mathbf{w} = (w_1, \dots, w_n)^T$, with $w_i = y_i \sigma(-y_i f_i)$, and $\mathbf{W} = \text{diag}[\sigma(y_1 f_1)(1 - \sigma(y_1 f_1)), \dots, \sigma(y_n f_n)(1 - \sigma(y_n f_n))]$, both computed at $\hat{\mathbf{f}}$. The predictive posterior distribution at a test set $\mathbf{X}_*$, see Eq 15.46, becomes

$$p(\mathbf{f}_*|\mathbf{X}_*, \mathbf{X}, \mathbf{y}) \approx N(\mathbf{f}_*; \mathbf{K}_*^T \mathbf{w}, \mathbf{K}_{**} - \mathbf{K}_*^T (\mathbf{K} + \mathbf{W}^{-1})^{-1} \mathbf{K}_*) \quad (15.50)$$

and $p(y_* = 1|\mathbf{X}_*, \mathbf{X}, \mathbf{y}) \approx \sigma(\mathbf{K}_*^T \mathbf{w})$.

Exercise

Derive Eq 15.50.

Hint. The Hessian in this case is $\mathbf{H} = -\mathbf{K}^{-1} - \mathbf{W}$, the iterated variance (see previous remark) leads to $\text{var}(\mathbf{f}_*) = \mathbf{K}_{**} - \mathbf{K}_*^T \mathbf{K}^{-1} \mathbf{K}_* + \mathbf{K}_*^T \mathbf{K}^{-1} \text{cov}(\mathbf{f}|\mathbf{y}) \mathbf{K}^{-1} \mathbf{K}_*$. Add to this the fact that $\text{cov}(\mathbf{f}|\mathbf{y}) = -\mathbf{H}^{-1}$ results in Eq 15.50 using again the Woodbury matrix identity.

Remark

Note that a less accurate approximation of the integral Eq 15.49 can be obtained by simply using $\int \sigma(x)N(x; \mu, \sigma^2)dx \approx \sigma(\mu)$ as in the above approximation of $p(y_* = 1|\mathbf{X}_*, \mathbf{X}, \mathbf{y})$.

15.5.3 Multi-class case

The method for the multi-class case, with say $\mathcal{C}_1, \dots, \mathcal{C}_C$, is similar to the binary case but using the softmax (or multinomial logistic) function. The softmax function $\boldsymbol{\sigma}()$, introduced in chapter 11, see also chapter 12 (section 3), maps the tuple $\mathbf{x} = (x_1, \dots, x_C)^T \in \mathbb{R}^C$ onto $[0, 1]^C$, $\boldsymbol{\sigma}(\mathbf{x}) = (\sigma_1, \dots, \sigma_C)^T$:

$$\sigma_c = \frac{e^{x_c}}{\sum_{l=1}^C e^{x_l}}, \quad c = 1, \dots, C. \quad (15.51)$$

As with the binary case we assume that each class is associated with a Gaussian process, i.e., $f_c \sim GP(0, K_c)$, $c = 1, \dots, C$, and can be assumed to be independent. That is, the prior for the c 'th class is:

$$p(\mathbf{f}_c|\mathbf{X}) \sim N(\mathbf{0}, \mathbf{K}_c), \quad c = 1, \dots, C \quad (15.52)$$

⁶it can also be written as $p(y|f) = \sigma(yf) = [\sigma(f)]^{\frac{1+y}{2}} [1 - \sigma(f)]^{\frac{1-y}{2}}$

where the covariance matrix $\mathbf{K}_c = (K_{c,ij})$ is obtained for example with the squared exponential using the test data input $\mathbf{X}$. The posterior for class $\mathcal{C}_c$, $c = 1, \dots, C$, conditional on the input sample $\mathbf{x}$, is:

$$p(\mathcal{C}_c|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_c)p(\mathcal{C}_1)}{\sum_{l=1}^C p(\mathbf{x}|\mathcal{C}_l)p(\mathcal{C}_l)} = \frac{e^{f_c(\mathbf{x})}}{\sum_{l=1}^C e^{f_l(\mathbf{x})}} \quad (15.53)$$

Now we consider the training set $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ with corresponding output (or response) $\mathbf{y} = (\mathbf{y}_1^T, \dots, \mathbf{y}_n^T)^T$ where each $\mathbf{y}_c = (y_{1c}, \dots, y_{nc})^T$ represents the labelling of the input $\mathbf{X}$ belonging to class $\mathcal{C}_c$, $c = 1, \dots, C$, i.e.,

$$y_{tc} = 1_{\mathcal{C}_c}(\mathbf{x}_t) = \begin{cases} 1 & \text{if } \mathbf{x}_t \in \mathcal{C}_c \\ 0 & \text{otherwise} \end{cases} \quad (15.54)$$

for $t = 1, \dots, n$ and $c = 1, \dots, C$. Following Eq 15.53 we let $\mathbf{p} = (\mathbf{p}_1^T, \dots, \mathbf{p}_C^T)^T$ where each $\mathbf{p}_c = (p_{1c}, \dots, p_{nc})^T$, $c = 1, \dots, C$, represents the probabilities of $\mathbf{x}_t \in \mathcal{C}_c$ for $t = 1, \dots, n$, i.e.,

$$p_{tc} = \frac{e^{f_c(\mathbf{x}_t)}}{\sum_{l=1}^C e^{f_l(\mathbf{x}_t)}} = \frac{e^{f_{tc}}}{\sum_{l=1}^C e^{f_{tl}}} \quad c = 1, \dots, C, \text{ and } t = 1, \dots, n. \quad (15.55)$$

The Gaussian prior of $\mathbf{f} = (\mathbf{f}_1^T, \dots, \mathbf{f}_C^T)^T$, with $\mathbf{f}_c = (f_{1c}, \dots, f_{nc})^T = (f_c(\mathbf{x}_1), \dots, f_c(\mathbf{x}_n))^T$, $c = 1, \dots, C$, contains the C latent functions and is Gaussian:

$$p(\mathbf{f}|\mathbf{X}) \sim N(\mathbf{0}, \mathbf{K}) \quad (15.56)$$

where $\mathbf{K} = \text{diag}(\mathbf{K}_1, \dots, \mathbf{K}_C)$ is a $Cn \times Cn$ bloc diagonal matrix containing the covariance matrices of the C classes. As with the binary case, see Eq 15.38, the probability of an input $\mathbf{x}$ to be in class $\mathcal{C}_y$ is:

$$p(\mathcal{C}_y|\mathbf{x}, \mathbf{f}) = \prod_{c=1}^C p(y_c = 1|\mathbf{x}, \mathbf{f})^{y_c} \quad (15.57)$$

where $y_c = 1_{\mathcal{C}_c}(\mathbf{x})$. Note that $(C - 1)$ factors among the C factors in Eq 15.57 are equal to 1. Its application to a sample $\mathbf{x}_t$, $t = 1, \dots, n$ is:

$$p(\mathcal{C}_{y_t}|\mathbf{x}_t, \mathbf{f}) = \prod_{c=1}^C p(y_{tc} = 1|\mathbf{x}_t, \mathbf{f})^{y_{tc}} \quad (15.58)$$

Posterior and predictive probabilities

The posterior probability $p(\mathbf{f}|\mathbf{X}, \mathbf{y})$ is computed in a similar fashion to the binary case, i.e., $p(\mathbf{f}|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{X})$. The last term is Gaussian, see Eq 15.58, and the likelihood $p(\mathbf{y}|\mathbf{f})$, see also Eqs 15.55 and 15.58, is given by:

$$p(\mathbf{y}|\mathbf{f}) = \prod_{t=1}^n \prod_{c=1}^C \left(\frac{e^{f_{tc}}}{\sum_{l=1}^C e^{f_{tl}}} \right)^{y_{tc}} \quad (15.59)$$

Note that for each t , $t = 1, \dots, n$, the vector $\mathbf{y}_c = (y_{t1}, \dots, y_{tC})^T$ contains $C - 1$ zeros and 1. This means that the products in Eq 15.59 contains only n terms, and $\sum_{c=1}^C y_{tc} = 1$. Since the likelihood is non-Gaussian, and like the binary case, Laplace approximation is again used to Gaussianize the posterior. The logarithm of the posterior is (up to a constant):

$$\mathcal{P}(\mathbf{f}) = \log p(\mathbf{y}|\mathbf{f}) + \log N(\mathbf{0}, \mathbf{K}) = \mathbf{y}^T \mathbf{f} - \sum_{t=1}^n \log \sum_{c=1}^C e^{f_{tc}} - \frac{1}{2} \log |\mathbf{K}| - \frac{1}{2} \mathbf{f}^T \mathbf{K}^{-1} \mathbf{f}, \quad (15.60)$$

which is to be approximated by a quadratic. The gradient $\nabla\mathcal{P}(\mathbf{f})$ is easily computed from Eq 15.60:

$$\nabla\mathcal{P}(\mathbf{f}) = \mathbf{y} - \mathbf{p} - \mathbf{K}^{-1}\mathbf{f}, \quad (15.61)$$

and the objective is to solve for $\nabla\mathcal{P}(\mathbf{f}) = \mathbf{0}$. The Hessian is:

$$\mathbf{H} = \nabla\nabla\mathcal{P}(\mathbf{f}) = \nabla(\mathbf{y} - \mathbf{p} - \mathbf{K}^{-1}\mathbf{f}) = -\mathbf{W} - \mathbf{K}^{-1}, \quad (15.62)$$

with $\mathbf{W}$ being the gradient of $\mathbf{p}$:

$$\mathbf{W} = \text{diag}(\mathbf{p}) - \mathbf{P}\mathbf{P}^T \quad (15.63)$$

where $\text{diag}(\mathbf{p})$ is the $Cn \times Cn$ diagonal matrix containing the elements of $\mathbf{p}$ in its diagonal, and $\mathbf{P}$ is a $Cn \times n$ matrix given by:

$$\mathbf{P} = \begin{pmatrix} \text{diag}(\mathbf{p}_1) \\ \vdots \\ \text{diag}(\mathbf{p}_C) \end{pmatrix} \quad (15.64)$$

Exercise

Derive the expression of $\mathbf{W}$ in Eq 15.63:

Answer. The derivative of the tj th, $t = 1, \dots, n; j = 1, \dots, C$, element of the probability vector $\mathbf{p}$ is $\frac{d}{df_{ic}} p_{tj} = \frac{de^{f_{tj}}/df_{ic}}{\sum_{l=1}^C e^{f_{tl}}} - \frac{e^{f_{tj}} d \sum_{l=1}^C e^{f_{tl}}/df_{ic}}{(\sum_{l=1}^C e^{f_{tl}})^2} = \frac{e^{f_{tj}}}{\sum_{l=1}^C e^{f_{tl}}} \delta_{it} \delta_{jc} - \frac{e^{f_{tj}} e^{f_{tc}}}{(\sum_{l=1}^C e^{f_{tl}})^2} \delta_{it}$. This is precisely $p_{tj} \delta_{jc} \delta_{ti} - p_{tj} p_{tc} \delta_{ti}$, for $t, i = 1, \dots, n; j, c = 1, \dots, C$. We can label the rows (and columns) of $\mathbf{W}$ by $11, 21 \dots n1, 12, 22 \dots n2, \dots, 1c, 2c \dots nc$ and refer to its elements by tj, ic indices. Now the first term in the above derivative is non-zero only when $t = i$ and $j = c$, and therefore represents $\text{diag}(\mathbf{p})$. The second term can be written as $p_{tj} p_{ic} \delta_{ti}$, in which the first two terms represent the elements of $\mathbf{p}\mathbf{p}^T$, but by including the term δ_{ti} , the only non zero terms correspond to $i = t$, and this corresponds precisely to replacing $\mathbf{p}$ in $\mathbf{p}\mathbf{p}^T$ by $[\text{diag}(\mathbf{p}_1) \text{diag}(\mathbf{p}_2) \dots \text{diag}(\mathbf{p}_C)]^T$ to yield $\mathbf{P}\mathbf{P}^T$.

Finally to get the Gaussian approximation of the posterior we iterate Newton's method as in the binary case, i.e. $\mathbf{f}_{k+1} = \mathbf{f}_k - \mathbf{H}^{-1} \nabla\mathcal{P}(\mathbf{f}_k) = (\mathbf{K}^{-1} + \mathbf{W})^{-1}(\mathbf{W}\mathbf{f}_k + \mathbf{y} - \mathbf{p})$. This iterative process converges to $\hat{\mathbf{f}}$, with associated Hessian $\hat{\mathbf{H}} = \mathbf{H}(\hat{\mathbf{f}})$, and then the normal approximation of the posterior is:

$$p(\mathbf{f}|\mathbf{X}, \mathbf{y}) \approx N(\hat{\mathbf{f}}, -\hat{\mathbf{H}}^{-1}) = N(\mathbf{K}(\mathbf{y} - \mathbf{p}), (\mathbf{K}^{-1} + \mathbf{W})^{-1}) \quad (15.65)$$

where the parameters of the above Gaussian distribution are evaluated at $\hat{\mathbf{f}}$.

Exercise

Show that in the iterative process $\mathbf{f}_k - \mathbf{H}^{-1} \nabla\mathcal{P}(\mathbf{f}_k) = (\mathbf{K}^{-1} + \mathbf{W})^{-1}(\mathbf{W}\mathbf{f}_k + \mathbf{y} - \mathbf{p})$.

Hint. From $-\mathbf{H}^{-1} \nabla\mathcal{P}(\mathbf{f}_k) = (\mathbf{W} + \mathbf{K}^{-1})^{-1}(-\mathbf{K}^{-1}\mathbf{f}_k + \mathbf{y} - \mathbf{p})$, expand the first term on the rhs, i.e., $(\mathbf{W} + \mathbf{K}^{-1})^{-1}(-\mathbf{K}^{-1}\mathbf{f}_k) = -\mathbf{W}^{-1}(\mathbf{W}^{-1} + \mathbf{K})^{-1}\mathbf{f}_k$, then use the Woodbury matrix identity (footnote 4 above) yields $-\mathbf{f}_k + (\mathbf{W} + \mathbf{K}^{-1})^{-1}\mathbf{W}\mathbf{f}_k$.

Now given a test input $\mathbf{X}_*$, the predictive posterior $p(\mathbf{f}_*|\mathbf{X}_*, \mathbf{X}, \mathbf{y})$ of $\mathbf{f}_*$ is estimated in a similar fashion to the binary case, see Eq 15.46 or Eq 15.50, and is.

$$p(\mathbf{f}_*|\mathbf{X}, \mathbf{X}_*, \mathbf{y}) \approx N(\mathbf{f}_*; \mathbf{K}_*^T(\mathbf{y} - \mathbf{p}), \mathbf{K}_{**} - \mathbf{K}_*^T(\mathbf{K} + \mathbf{W}^{-1})^{-1}\mathbf{K}_*) \quad (15.66)$$

Recalling that $\mathbf{K}$ is bloc-diagonal, due to the uncorrelation between the classes, the above predictive posterior mean can be separated into K means, $\mathbf{K}_{*,c}^T(\mathbf{y}_c - \mathbf{p}_c)$, $c = 1, \dots, C$. For a particular test input $\mathbf{x}_*$ the predictive posterior can be used to obtain the probability $\pi_{*,c} = p(\mathcal{C}_c|\mathbf{x}_*)$,

$$\pi_{*,c} = \frac{e^{\mu_{*,c}}}{\sum_{l=1}^C e^{\mu_{*,l}}}, \quad c = 1, \dots, C \quad (15.67)$$

where $\mu_{*,c} = \mathbf{k}_{*,c}^T(\mathbf{y}_c - \mathbf{p}_c)$, for $c = 1, \dots, C$. The correct class of $\mathbf{x}_*$ is the one corresponding to the maximum of $\{\pi_{*,1}, \dots, \pi_{*,C}\}$.

15.6 Learning the kernel and hyperparameters

15.6.1 Choice of the covariance function

There are different types of kernels used in GP modelling. The standard (stationary) covariance function GP regression or classification is the Gaussian kernel, also known as the squared exponential (or radial basis function) is

$$K(\mathbf{x}_i, \mathbf{x}_j) = \theta_1 \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\theta_2}\right) \quad (15.68)$$

The parameters θ_1 and θ_2 control the amplitude and smoothness of the process. Large θ_2 generates smooth paths and vice-versa for small values. Note that $\sqrt{\theta_2}$ gives a measure of variability of the process. Another example is the exponential kernel, $K(\mathbf{x}_i, \mathbf{x}_j) = \theta_1 \exp(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|}{\theta_2})$. This corresponds to the Ornstein-Uhlenbeck process (Uhlenbeck and Ornstein 1930) to describe a Brownian motion, and is therefore rougher than the squared exponential. Other kernels with "non-stationary" covariances can also be used like the scalar product $K(\mathbf{x}_i, \mathbf{x}_j) = \theta_1 + \theta_2 \mathbf{x}_i^T \mathbf{x}_j$. Several other choices can be endorsed, for example through addition or convolution, see Lindholm et al. (2022) or Bishop (2006). General formulation of covariance function of stationary Gaussian process can be obtained using Bochner theorem. The theorem represents these covariances via Fourier transform of positive semi-definite function (chapter 2), see Rasmussen and Williams (2006, chap. 4) for further details.

Remark

The choice of kernels can be guided by the smoothness or roughness of the phenomenon at hand. For example, Gaussian kernels are in general smoother than exponential kernels as shown in the examples discussed below.

Figure 15.3 shows three cases of 10 samples each of a Gaussian process with $\theta_1 = 1/\sqrt{4\pi}$ and θ_2 equals to 10 (a), 0.2 (b) and 0.01 (c). Note that a weak noise was added to the observations. Figure 15.4 is similar to Figure 15.3 but for an exponential kernel with same values of hyperparameters θ_1 and θ_2 .

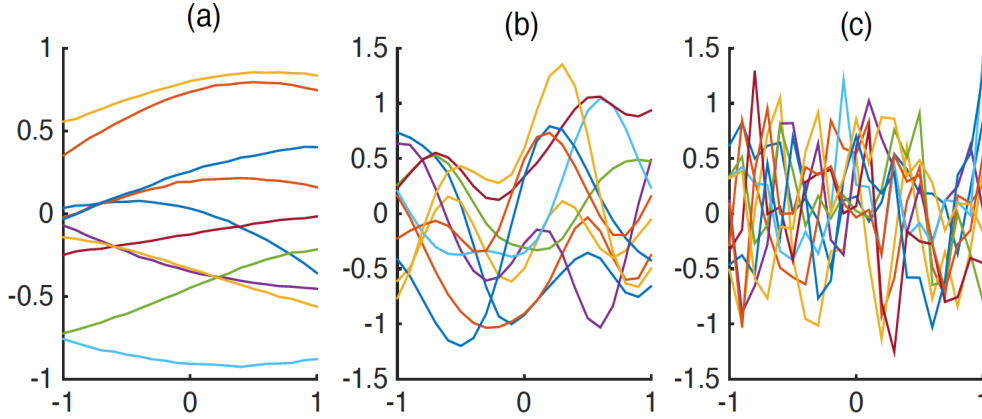


Figure 15.3: Samples of GP paths using Gaussian covariance function (Eq 15.68) with parameter $\theta_1 = 1/\sqrt{4\pi}$, and θ_2 equals 10 (a), 0.2 (b), and 0.01 (c). A weak Gaussian noise is added to the observations.

Figure 15.5 shows an example of a one-dimensional GP fitting and prediction. The covariance matrix is based on a Gaussian kernel. The blue curve represents the function $y(x) = 1 + \frac{x}{20} + \frac{\sin x}{x}$. Eight inputs x_t , $t = 1, \dots, 8$, are randomly generated, and to fit a GP regression model observations, $y_t = y(x_t) + \varepsilon_t$, are constructed (shown by small circles), with Gaussian iid ε_t . Then 100 equidistant points in $[0, 14]$ are used as inputs $\mathbf{X}_*$. The predictive posterior of $\mathbf{f}_*$ (or $\mathbf{y}_*$) is given by Eq 15.20 and its mean is shown by the red curve. The shading shows two standard deviations around the mean. Note the small uncertainty near the observations. Also, unlike other fitting models there are infinite possible posterior functions or paths that can be drawn from the predictive posterior. To generate one path, one just create a vector (with dimension 100 like $\mathbf{f}_*$) from the predictive posterior Eq 15.20. The following is a basic Matlab code to generate the figure.

```
x=(5*randn(9,1)+10);
y=1+x*5e-2+sin(x)./x+0.1*randn(9,1);
xn=(linspace(0,14,100))'; yn=1+xn*5e-2+sin(xn)./xn;
gpr=fitrgp(x,y);
[ypred ystd yInt]=predict(gpr, xn);
patch([xn; flipud(xn)], [ypred-ystd; flipud(ypred+ystd)], 'b'); hold on
plot(xn, yn, 'b'); plot(xn, ypred, 'r'); plot(x, y, 'ko');
```

Note that the above code will not generate exactly the same figure because of the random components.

15.6.2 Hyperparameters estimation

The hyperparameters of the covariance function $K()$ have to be estimated based on the training dataset. From a Bayesian framework perspective the posterior distribution for model

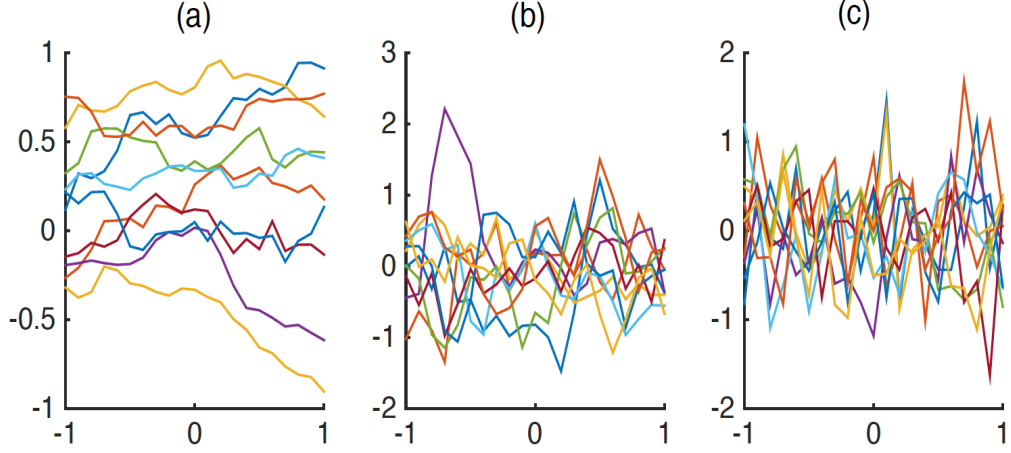


Figure 15.4: As in Fig 15.3 but using an exponential covariance function with similar amplitude and scale parameters.

parameters $\boldsymbol{\theta} = (\theta_1, \dots, \theta_M)$, given the data $\mathbf{y}$ is

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}) \quad (15.69)$$

Assuming flat prior distribution, if the posterior distribution is peaked around $\hat{\boldsymbol{\theta}}$, then these parameters are obtained by maximizing $p(\mathbf{y}|\boldsymbol{\theta})$, which is known as the *marginal likelihood function* or *model evidence*, see Rasmussen and Williams (2006) or Bishop (2006) for the reasoning behind this concept.

Parameter estimation of GP regression model

For the GP regression model the log-likelihood, see Eq 15.21, i.e. $\log(p(\mathbf{y}|\mathbf{X})) = \log(N(\mathbf{y}; \mathbf{0}, \mathbf{K}))$, and reads $\log(p(\mathbf{y}|\boldsymbol{\theta})) = -\frac{1}{2}(\log |\mathbf{K}| + \mathbf{y}^T \mathbf{K}^{-1} \mathbf{y}) + \text{const.}$ Its gradient (needed for the optimization) reads

$$\frac{\partial}{\partial \theta_i} = -\frac{1}{2} \text{tr}(\mathbf{K}^{-1} \frac{\partial}{\partial \theta_i} \mathbf{K}) + \frac{1}{2} \mathbf{y}^T \mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta_i} \mathbf{K}^{-1} \mathbf{y} = \frac{1}{2} \text{tr} \left(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta_i} \mathbf{K}^{-1} (\mathbf{y} \mathbf{y}^T - \mathbf{K}) \right) \quad (15.70)$$

If the training data is noisy, see Eq 15.21, then $\mathbf{K}$ in the above equation is replaced by $\mathbf{K} + \sigma^2 \mathbf{I}$ where the variance σ^2 is assumed to be known. During the optimization of the log-likelihood, most of the computation time is used to compute $\mathbf{K}^{-1}$, which is $O(n^3)$, based on Cholesky decomposition. The above gradient depends also on the type of the covariance function and the number of parameters.

Parameter estimation of GP classification model

The likelihood of the (binary) classification case is different from that of the GP regression because of the logistic sigmoid, compare Eqs 15.16 and 15.42. The likelihood is given by

$$p(\mathbf{y}|\boldsymbol{\theta}) = \int p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\boldsymbol{\theta})d\mathbf{y} \quad (15.71)$$

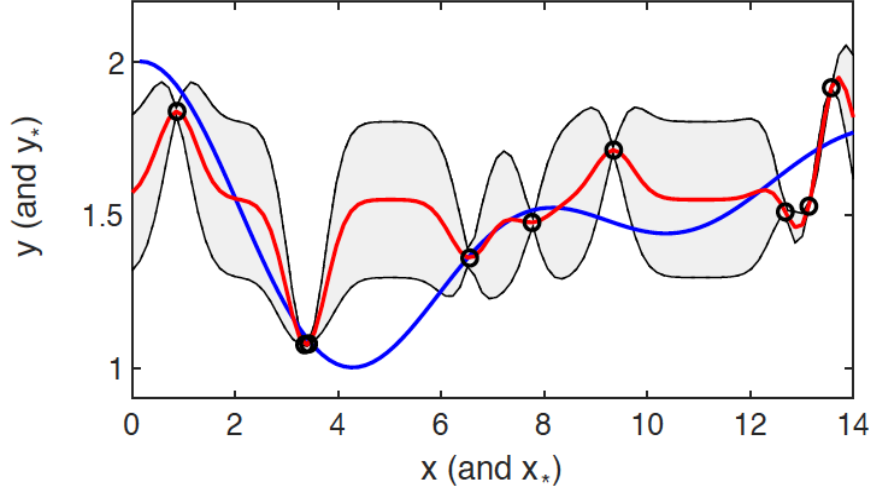


Figure 15.5: Example of GP regression model using eight observations generated from the curve $y(x) = 1 + \frac{x}{20} + \frac{\sin x}{x}$ (blue) contaminated with a Gaussian noise (small circles). The mean predictive distribution is shown by the red curve, and the shading shows two standard deviations around this mean.

where $p(\mathbf{y}|\mathbf{f})$ is given by Eq 15.41 and $p(\mathbf{f}|\boldsymbol{\theta}) \sim N(\mathbf{0}, \mathbf{K})$. Equation 15.71 is again computed using Laplace approximation. The log-likelihood $\log p(\mathbf{y}|\boldsymbol{\theta})$ is approximated (up to a constant term) by $\log p(\hat{\mathbf{f}}) + \log p(\mathbf{y}|\hat{\mathbf{f}}) - \frac{1}{2} \log |\mathbf{K}^{-1} + \mathbf{W}|$, which boils down to

$$\log p(\mathbf{y}|\boldsymbol{\theta}) = \mathbf{y}^T \hat{\mathbf{f}} - \sum_{i=1}^n \log(1 + e^{\hat{f}_i}) - \frac{1}{2} \hat{\mathbf{f}}^T \mathbf{K}^{-1} \hat{\mathbf{f}} - \frac{1}{2} \log |\mathbf{K}| - \frac{1}{2} \log |\mathbf{K}^{-1} + \mathbf{W}| \quad (15.72)$$

Note that since $\sigma(yf)(1 - \sigma(yf)) = \sigma(f)(1 - \sigma(f))$, for $y \in \{1, -1\}$, the matrix $\mathbf{W}$ becomes $\mathbf{W} = \text{diag}[(\sigma(f_1)(1 - \sigma(f_1)), \dots, \sigma(f_n)(1 - \sigma(f_n))]$. The gradient of Eq 15.72 is again required for the optimization. Note also that $\mathbf{W}$ and $\hat{\mathbf{f}}$ vary with the hyperparameters $\boldsymbol{\theta}$, i.e. changes in the hyperparameters will lead to changes in $\mathbf{W}$ and $\hat{\mathbf{f}}$. Note also that $\log |\mathbf{K}| + \log |\mathbf{K}^{-1} + \mathbf{W}| = \log |\mathbf{I} + \mathbf{K}\mathbf{W}|$, which can be transformed to yield $\log |\mathbf{I} + \mathbf{W}^{1/2} \mathbf{K} \mathbf{W}^{1/2}|$ involving a symmetric positive matrix.

Exercise

Show that $\log |\mathbf{K}| + \log |\mathbf{K}^{-1} + \mathbf{W}| = \log |\mathbf{I} + \mathbf{W}^{1/2} \mathbf{K} \mathbf{W}^{1/2}|$.

(Use the identity: $\mathbf{K}(\mathbf{K}^{-1} + \mathbf{W}) = \mathbf{W}^{-\frac{1}{2}}(\mathbf{I} + \mathbf{W}^{\frac{1}{2}} \mathbf{K} \mathbf{W}^{\frac{1}{2}}) \mathbf{W}^{\frac{1}{2}}$).

15.7 Software resource material

- Matlab-R2024 has a script "fitgpr" that performs fitting and prediction using GP regression model to a given dataset.
- The textbook by Wang (2026) has essential Matlab scripts of GP regression and classification.

- Python scripts are available online that perform GP regression and classification, see for the example

<https://nbviewer.org/github/krasserm/bayesian-machine-learning/tree/dev/gaussian-processes/>